“Develop a passion for learning.”

z/OS Job Control Language (JCL)

JCL STATEMENT FORMAT AND CODING RULES

TEKsystems Global Services

WORKFORCE DEVELOPMENT

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GENERAL JCL STATEMENT FORMAT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name   operation parameters comments                                 Csequence
```

Identifier field **columns 1-2**

The value of 2 slashes identifies the record as a JCL statement.

If the third position contains an asterisk and the fourth position contains either another asterisk or a space, the whole record is a JCL comment.

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GENERAL JCL STATEMENT FORMAT

```
          1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name    operation  parameters  comments                                Csequence
```

Name field   starts in column 3

If the statement has a name, the value begins in column 3 and can be up to eight (8) characters long.

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GENERAL JCL STATEMENT FORMAT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name   operation parameters comments                               Csequence
```

Operation field **one or more spaces after name**

Every JCL statement (except comments) has an operation.

- JOB – the statement marks the beginning of a job.
- EXEC – the statement defines a job step.
- DD – the statement describes a file used in the job step.

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GENERAL JCL STATEMENT FORMAT

```
          1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name    operation parameters comments                                 Csequence
```

Parameter field   **one or more spaces after operation**

Each JCL operation can take optional parameters that give JES specific information about how to process the statement.

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GENERAL JCL STATEMENT FORMAT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name    operation  parameters  comments                                    Csequence
```

Comment field **after parameters, up to column 72**

Anything following the parameters and a blank space is treated as a comment, through column 71 (not including 72).

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GENERAL JCL STATEMENT FORMAT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name   operation  parameters  comments                                 Csequence
```

Continuation character column 72

Any non-blank character in column 72 tells JES the following line is a continuation of the current line.

Often, the space available in the 80-column record is insufficient for all the information on a JCL statement. In those cases, we can continue the JCL statement on the next line by marking the current line with a non-blank character in column 72.

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GENERAL JCL STATEMENT FORMAT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//name    operation  parameters  comments                                Csequence
```

Sequence field **columns 73-80**

Originally used for a sequence number, this field is not usually used anymore. This was useful when JCL statements were punched into cards. The deck could be dropped on the floor or otherwise disordered, and then re-ordered by a sorting machine.

The field can contain any data we want – coding something here will do no harm – but it is rarely used.

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SAMPLE JOB STREAM

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*****************************************************************
//* This job does some really cool stuff.
//*****************************************************************
//MYJOB1 JOB . . .
//* This step clears out files that may be left over from previous runs
//STEP1  EXEC . . .
//* This file cannot be empty when the program runs
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3   DD . . .
```

Let's take a look at this sample job stream and identify which lines are JCL statements and which are not.

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JCL STATEMENTS IDENTIFIED BY SLASHES IN COLUMNS 1-2

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*********************************************************************
//* This job does some really cool stuff.
//*********************************************************************
//MYJOB1 JOB . . .
//* This step clears out files that may be left over from previous runs
//STEP1  EXEC . . .
//* This file cannot be empty when the program runs
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3   DD . . .
```

The lines that begin with slashes in columns 1 and 2 are JCL statements.

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JCL COMMENT LINES

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*****************************************************************
//* This job does some really cool stuff.
//*****************************************************************
//MYJOB1 JOB . . .
//* This step clears out files that may be left over from previous runs
//STEP1  EXEC . . .
//* This file cannot be empty when the program runs
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3  DD . . .
```

These lines do not begin with two slashes. JES treats them as data.

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JCL COMMENT LINES

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*********************************************************************
//* This job does some really cool stuff.
//*********************************************************************
//MYJOB1 JOB . . .
//* This step clears out files that may be left over from previous runs
//STEP1  EXEC . . .
//* This file cannot be empty when the program runs
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3   DD . . .
```

Lines that begin with two slashes and an asterisk are probably comments, but there's one more thing to check to identify a comment...

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JCL COMMENT LINES

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*******************************************************************
//* This job does some really cool stuff.
//*******************************************************************
//MYJOB1 JOB . . .
//* This step clears ou
//STEP1  EXEC . . .
//* This file cannot be
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3   DD . . .
```

If the installation is running JES3, then lines beginning with slash, slash, asterisk, non-blank character in columns 1-4 are JES3 commands.

JCL comment lines begin with slash, slash, asterisk, followed by a blank space or another asterisk.

RULES FOR NAMES

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//X       JOB . . .                OK – 1 character from ISO8859-1
//ABCDEFGH JOB . . .               OK – 8 characters from ISO8859-1
//$FOO    JOB . . .                OK – ISO 8859-1 and $
//#ABC@123 JOB . . .               OK – ISO 8859-1 and # and @
//税金計算   JOB . . .               No – Unicode not allowed here
//DANĚ JOB . . .                   No – ISO 8859-2 not allowed here
// ARC55 JOB . . .                 No – there is a space after the 2 slashes
//XYZ66JOB . . .                   No – there is no space between name and operation
```

- Must begin in column 3 immediately after the 2 slashes.
- Must be between 1 and 8 characters long.
- Must be followed by a space before the operation value.
- Can include characters from character set ISO 8859-1, EBCDIC…
- …plus the characters $, #, and @.

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NULL STATEMENT

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890

//*********************************************************************
//* This job does some really cool stuff.
//*********************************************************************
//MYJOB1 JOB . . .
//* This step clears out files that may be left over from previous runs
//STEP1  EXEC . . .
//* This file cannot be empty when
//FILE1  DD . . .
//FILE2  DD *
Data record one
Data record two
/*
//STEP2 EXEC . . .
//FILE3   DD . . .
//
```

Optionally, a job stream can end with a *null statement* consisting of just the two slashes in columns 1 and 2, with the rest of the record blank.

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RESERVED DD STATEMENT NAMES

For the name field on the Data Definition statement, known as the DD Name, some names are reserved for system use. We cannot use these names for our own data sets.

A list of reserved DD Names and an explanation of how the system uses them are available at this URL:

```
https://www.ibm.com/docs/en/zos-basic-skills
 ?topic=jdsausn-jcl-dd-statement-ddnames-that-are-reserved-specific-uses
```

LAB 11: IDENTIFY JCL STATEMENTS

Follow the instructions for Lab 11 in labs/labs.md in the course repo.

WHAT YOU KNOW AND WHAT YOU CAN DO

- You know the general format of JCL statements, and the historical reason for that format.
- You can recognize JCL statements vs. other records in a JCL file, and you can identify the fields on each statement.

WHAT'S NEXT

- The details of the Job statement